

Sample Exam: 1

Section 1: Conceptual Understanding

Data Structures: Explain the core difference between a List, a Tuple, and a Set. Provide one specific business scenario for each where it would be the absolute best choice to use.

Functions & Modularity: What is the difference between `print()` and `return` inside a function? Why is using `return` essential when building reusable systems like a payroll calculator?

Scope: Explain the difference between a local variable and a global variable. In what scenario would you need to use the `global` keyword inside a function?

Computational Thinking: Name the four pillars of computational thinking. Take a real-world business problem (like managing a restaurant's inventory) and briefly describe how you would apply two of those pillars to solve it.

OOP Concepts: Define "Inheritance" and "Polymorphism" in Object-Oriented Programming. Why are these concepts useful when writing software for a company?

Data Analysis (EDA): Why is Exploratory Data Analysis referred to as a step to prevent "Garbage in = Garbage out"? Name two things you must check for during EDA before using data for machine learning.

File Handling: What is the difference between opening a file in 'w' (write) mode versus 'a' (append) mode? What happens to the existing data in the file in both cases?

Exception Handling: What is a runtime error? Why is it better to use a `try-except` block rather than just letting the program crash?

Section 2: Syntax and Code Tracing

Function Arguments: What is the strict rule for the order of parameters when defining a function? (Where do required parameters, default parameters, `*args`, and `kwargs` go in the `def` statement?)

Syntax Correction: Identify the syntax errors in the following class definition and write the corrected version.

```
class Employee
    def __init__(name, salary):
        self.name = name
        salary = salary
```

Code Tracing (Mutability): What will be the output of the following code? Explain why this happens based on the rules of mutable vs. immutable data types.

```
def update_price(price):
    price = price + 500

item_price = 1000
update_price(item_price)
print(item_price)
```

Dictionary Syntax: Write the exact Python syntax to create a dictionary that maps three Employee IDs (101, 102, 103) to their respective salaries. Then, write the line of code to look up and print the salary of Employee 102.

Tkinter Syntax: Write the basic syntax structure (about 4-5 lines of code) required to create an empty Tkinter Graphical User Interface (GUI) window that stays open on the screen.

Section 3: Short Application Scenarios

Decision Structures: A company gives a bonus based on sales and attendance. If an employee's sales are above 100,000 AND their attendance is above 90%, they get a bonus. If their sales are above 100,000 but attendance is lower, they get half the bonus. Write a nested if or if-elif-else structure to represent this logic.

OOP Initialization: You are designing a system for a bank. Write a base class called Account with an `__init__` method that sets a balance. Then, write a child class called SavingsAccount that inherits from Account, adds an `interest_rate` attribute, and uses the `super()` function correctly to initialize the parent.

Pandas/NumPy Application: Assume you have a Pandas DataFrame named `df` containing employee data with columns `['Name', 'Department', 'Salary']`. Write the exact code you would use to:

- Find and fill any missing values in the 'Salary' column with the mean salary.
- Group the DataFrame by 'Department' to find the total sum of salaries per department.

File Reading & Loops: Write a small script using the with open(...) structure to open a text file called transactions.txt in read mode and print each line one by one using a for loop.

Sample Exam: 2

Section 1: Conceptual Understanding

The IPO Model: Using an ATM cash withdrawal as an example, explicitly identify what constitutes the Input, the Processing, and the Output.

Data Structure Selection: You are building a system to store product prices for a supermarket (e.g., Apple: \$2, Banana: \$1). Explain why a Dictionary is the perfect choice for this, and explain specifically why a List and a Set would fail or be highly inefficient for this specific task.

Loops: What is the fundamental difference between a for loop and a while loop regarding when you choose to use them? What is the specific risk associated with while loops that does not exist in for loops, and how do you prevent it?

OOP Access Modifiers: Explain the difference between Public, Protected (`_`), and Private (`__`) attributes in Python classes. Explain the concept of "Name Mangling" and how Python uses it to enforce Private variables.

Method Overriding vs. Overloading: In Object-Oriented Programming, what is the difference between Method Overriding and Method Overloading? Which one occurs within the same class, and which one requires inheritance?

Exploratory Data Analysis (EDA): Describe the "Five-Number Summary" used to create a box plot. What specific data issue does a box plot help you visually detect?

NumPy vs. Standard Python: What does the term "Vectorization" mean in the context of NumPy? Why is NumPy preferred over standard Python lists when doing numerical computations for Machine Learning?

Tkinter Layouts: Compare the `pack()`, `grid()`, and `place()` layout managers in GUI programming. Briefly explain the mechanism of how each one organizes widgets on the screen.

Section 2: Syntax and Code Tracing

Loop Tracing: What is the exact visual output of the following nested loop?

```
for i in range(3):
    for j in range(i + 1):
        print('*', end='')
    print()
```

Exception Syntax: Write a complete try-except-finally block that attempts to open a file called data.txt in read mode. If it fails due to the file not existing, it should catch a FileNotFoundError and print "File missing". Regardless of whether it succeeds or fails, it should print "Process finished" at the very end.

OOP Encapsulation Trace: Look at the following code. Why does the last line cause an error? Write the exact line of code that would successfully print the marks (using the mangled name).

```
class Student:
    def __init__(self, name):
        self.name = name
        self.__marks = 80

s1 = Student("Ali")
print(s1.name)
print(s1.__marks)
```

Default Arguments Trace: What are the two values printed by this code? Explain how the default parameter dictates this behavior.

```
def calculate(a, b=5):
    return a * b

print(calculate(10))
print(calculate(10, 2))
```

Section 3: Short Application Scenarios

List Operations: A retail store has their daily sales stored in a list: daily_sales = [5000, 7000, 6500]. Write the Python code to append a new day's sale of 8000 to this list, and then write the code to calculate and print the total sum of all the sales in the updated list.

Decision Logic: You are writing a login system. You have two boolean variables: is_registered and password_correct. Write the if-elif-else logic to:

- Print "Login Successful" if both are true.

- Print "Invalid Password" if they are registered but the password is wrong.
- Print "Please Register" if they are not registered at all.

OOP Inheritance: Create a base class called Employee with a method get_salary() that simply returns 50000. Create a child class called Manager that inherits from Employee but overrides the get_salary() method to return 80000.

Pandas Application: Assume you have a Pandas DataFrame named df containing the columns ['Product', 'Category', 'Price']. Write the Python code to:

- Filter the DataFrame to select only the rows where the 'Price' is greater than 1000.
- Use the groupby() function on that filtered data to find the mean() price for each 'Category'.